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Posted Date: 12 October 2026

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Article

Projection-Defined Physicality in PT-Symmetric Quaternionic Spacetime

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Abstract

Generalized geometric frameworks can admit enlarged kinematical structure without thereby specifying a physical observable sector. Normalization, expectation values, conserved transport, and admissible evolution then require an explicit selection rule. This paper formulates PT-symmetric quaternionic spacetime (PTQ) under the Observable Branch Principle (OBP). OBP is an organizing principle of this program: PTQ is not asserted to be the unique microscopic realization of every possible theory, but a projection-defined framework for an admissible observable sector. We state the minimal principles adopted by that framework and separate them from results that are standard elsewhere, from conditional constructions in companion papers, and from open problems. A positive physical metric, a probability interpretation, and a local conserved current are not consequences of projection idempotency alone. They require a declared complex structure, a compatible positive metric, an observable representation, probability-measure hypotheses, and dynamics with whatever additional locality is needed for a local current. A finite-dimensional toy model makes these distinctions explicit. The paper does not present a cosmological model, does not replace quantum theory, does not derive a universal closed dynamical system, and does not claim that one residual geometry has already been identified across observational regimes.

Keywords: PT-symmetric quaternionic spacetime; observable branch principle; projection-defined observability; physical complex structure; pseudo-Hermitian quantum mechanics; projected dynamics

1. Introduction

A PT-symmetric quaternionic framework can possess enlarged kinematical structure, while empirical physics is formulated only for quantities that can be identified, compared, and normalized. Without a distinction between formal structure and observable content, kinematical degrees of freedom are easily read as physically reportable ones. The manuscript is organized around the following claim, adopted as a principle of PTQ rather than derived as a theorem about every generalized quantum theory: in this program, observable physicality is fixed only after projection onto an admissible observable sector.

That claim is the Observable Branch Principle (OBP). The quaternionic geometry considered here is not assumed to be the unique microscopic realization of all possible universes. It is proposed as the effective geometry of a PT-even observable sector obtained by admissible projection, or by an explicitly declared coarse graining, from an enlarged configuration space. In this paper a branch is an admissible observable sector. The word is not a many-worlds ontological claim, and OBP is not a theorem that every physical theory must use the same selector.

The manuscript is not a reformulation of PT-symmetric quantum mechanics and not a generic non-Hermitian extension of ordinary quantum theory. In those settings, physical admissibility is commonly posed for operators together with a metric that makes them acceptable [1–4]. PTQ adopts a different order for its own framework: a candidate observable sector is declared by projection, and only then are a complex structure, a positive metric, a probability interpretation, and observable dynamics additional data on that sector. This order is a choice of organization. It does not show

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that operator-level metric selection is inconsistent, and it does not show that projection is the unique resolution available to every generalized theory.

Modified-gravity constructions often organize departures at the level of the low-energy gravitational sector [5,6]. When PTQ is used to interpret an observable deviation, the intended reading is that the deviation belongs to projection-induced residual structure in a declared sector. That reading is a proposal about attribution. It is not a proof that a unique constitutive model has been selected, and it is not an observational validation.

The paper states the principles OBP uses, explains what those principles do and do not imply, records the conditional probability and continuity requirements without reproducing the companion probability theorems, presents projected dynamics as a class of constraints, and states conservative failure conditions. It does not re-prove tensor-sector identities. It does not reproduce a projective classification. It does not present weak-field fits, likelihood analyses, or collider templates. It does not claim that one partial differential equation governs all scales.

Companion papers supply conditional results in declared scopes: tensor-sector identities under stated assumptions, a scalar-operator comparison of strict and Schur reduction, a probability-equivalence analysis, a local trace-residue classification, and separate phenomenological or geometric branches. Section 7 states those scopes. The branches are not joined here by one universal theorem.

The result of the paper is therefore limited. OBP is the organizing principle. The only nontrivial mathematical statement proved here is the toy-model proposition of Section 5.6: a real-linear PT-even projection need not be complex-linear for the ambient complex structure, but it can be complex-linear for a declared complex structure J, and a chosen J-Hermitian product can restrict to the standard amplitude product on the image. That proposition does not fix a unique metric for PTQ, does not derive Born's rule, and does not produce a local current from norm conservation alone.

2. Minimal Principles of PTQ

The framework uses four principles. They are the organizing assumptions of this paper. The text does not prove that no weaker set could serve some other theory, and it does not prove that these four principles are forced by physical law independently of the decision to adopt OBP.

2.1. Principle I: Quaternionic Kinematical Structure

PTQ begins from the assumption that the underlying kinematical description is quaternionic. Admissible kinematics, in this posture, are those for which an involutive PT operation belongs to the architecture of the arena. The principle states the arena in which the later projection problem is posed. Without an enlargement of kinematical structure, that problem would not be the one this paper is about. The assumption is not itself a derivation of quaternionic spacetime from a more general physical law.

2.2. Principle II: Projection-Defined Observability

Physical observables are not identified with the full quaternionic kinematical space. Observable content is defined only after projection onto a sector that the framework counts as admissible. A quantity that exists kinematically but does not survive that projection is not part of the observable content.

Admissibility, as used in this manuscript, imposes four requirements on a candidate projection. The image must be closed as the domain in which observables are compared. It must admit a declared complex structure and at least one positive inner product compatible with the norms and expectation values one intends to use. Admissible evolution, once a dynamical law is specified, must preserve the sector. The sector must be able to carry the probability and continuity structure required by the interpretation one attaches to it. These are entrance conditions. They do not classify all projections, they do not prove that a projection exists in every geometric model, and they do not select a unique sector.

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2.3. Principle III: Compatible Physical Inner Product

The physical inner product is not given a priori on the full kinematical space, and it is not fixed by idempotency of the projection. Once a sector and a complex structure on that sector have been declared, a positive inner product used for norms and expectation values must be Hermitian with respect to that complex structure and compatible with the observables retained there. In pseudo-Hermitian and quasi-Hermitian quantum mechanics it is already standard that a positive metric making a generator Hermitian need not be unique, and that a larger observable algebra can constrain it further [2–4]. OBP inherits that distinction. The toy model below exhibits one compatible product and the special sense in which its scale is conventional on a one-dimensional complex line. That example is not a uniqueness theorem.

2.4. Principle IV: Projected Observable Dynamics

Physically relevant dynamics, in this framework, are dynamics of the projected observables, not the unrestricted motion of the full kinematical structure read as directly physical. The underlying evolution may be richer. Empirical dynamics are the induced evolution on the admissible sector, and only when that evolution preserves the sector, the chosen metric, and the continuity structure one has claimed. This paper does not adopt a single effective equation for all regimes. Companion results that constrain particular geometric sectors are cited only inside the scopes recorded in Section 7.

3. Projection-Defined Physicality in the PTQ Posture

The role of projection in PTQ can be stated as a three-step posture. First, identifying the full formal space with the observable sector fails for the enlarged kinematics this program uses. Second, enlarging the algebra does not by itself name the physical sector, the physical complex structure, or the physical metric. Third, OBP adopts admissible projection as the organizing selection rule. The third step is a principle of this framework. It is not shown here to be the only possible selection rule in generalized quantum theory.

3.1. Failure of Naive Identification

If the full kinematical space were identified with the physical sector, every formally available degree of freedom would be treated as reportable. That identification is harmless only when the kinematical arena and the observable arena are already the same. PTQ does not start from that situation. A generalized state space can contain components whose algebraic role is part of the formulation and whose direct physical meaning is still unsettled. Reading all of them as observables leaves reality conditions, expectation values, and the physical significance of evolution unfixed.

3.2. Why Extension Alone Is Insufficient

An extension of the formal arena does not uniquely define the physical sector. It does not single out the inner product used to interpret observables. It does not by itself enforce the reality conditions imposed on those observables. The same point applies in quaternionic form. A larger algebra can support inequivalent readings until a sector, a complex structure, and a compatible positive metric are supplied. Those data are not postponed to applications, because any probability or current statement made by the framework already depends on them.

3.3. What OBP Adopts, and What It Does Not Replace

OBP requires that physical quantities be defined by survival under an admissible projection from the enlarged arena to a sector on which interpretation is possible. After that declaration, a physical inner product is a metric compatible with the sector, not an arbitrary form on the full kinematics. Probability and current, if they are introduced, are structures on that same sector and under the further hypotheses of Section 4. Dynamics are then required to preserve the declared interpretation.

This is not a claim that projection replaces every other restriction mechanism. Constrained quantization, superselection, pseudo-Hermitian metric selection, Schur–Feshbach elimination, and

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projective gauge quotient are standard and distinct operations. Section 7.3 separates the ones that appear in the public PTQ papers. OBP does not derive those operations from one another, and it does not assert that a theory which uses one of them must also use OBP.

3.4. Projection as the PTQ Criterion of Observable Physicality

Inside PTQ, projection is the criterion that marks the boundary between kinematical data and the observable sector. Without some such criterion the program would have enlarged kinematics and no declared observable domain. That is a statement about the intelligibility of this program. It is not a proof that projection is a technical necessity in every generalized quantum theory, and it is not a proof that the criterion determines a unique sector or a unique metric. The later sections use the criterion only in that limited sense.

4. Conditional Probability and Continuity Structure

This section states the dependency that a probability interpretation has inside OBP. It does not postulate a probability rule on unrestricted quaternionic kinematics, and it does not derive one from projection idempotency. The detailed equivalence and obstruction theorems belong to the companion study of pseudo-Hermitian probability on supplied admissible sectors [7]. They are not reproduced here. The two papers have different jobs: this one fixes the foundational order in which PTQ is willing to talk about probability; that one studies when supplied projected realizations represent the same probability theory.

4.1. Data Required on an Admissible Sector

A conditional physical probability statement, in the sense used here, presupposes all of the following.

1. An enlarged kinematics and a declared admissible projection, or an equivalent declared retention rule, selecting a sector.
2. A complex structure on that sector, relative to which the physical Hilbert-space operations are defined. In the toy model this is the structure J of Section 5.6, not the ambient complex structure.
3. A positive metric G, Hermitian with respect to that complex structure, compatible with the retained generator and with the represented observables. Idempotency of the projection does not determine G.
4. An observable representation on the sector, as distinct from the mere selection of a state subspace.
5. Probability-measure hypotheses and preparation calibration. Where a Gleason-type representation is invoked, the companion paper states the dimension, additivity, and noncontextuality hypotheses explicitly and excludes complex dimension two. Those hypotheses are inputs. A conserved norm does not supply them.
6. Dynamics compatible with the metric on the stated domain. A local conserved current is a further statement and needs locality and kinetic, or other transport, structure. Global norm conservation does not imply a model-independent local continuity equation.

Under those hypotheses one may speak of a conditional probability interpretation on the declared sector. The list does not provide a probability measure on the labels of the projections themselves, does not prove that every admissible sector is operationally equivalent to every other, and does not solve the measurement problem.

4.2. Distinctions Preserved

The following are different assertions, and this manuscript does not collapse them. Physical-sector selection is the declaration of the sector. Metric compatibility is the choice of a positive G appropriate to that sector and to its observables. Probability normalization is a condition on states in the G-inner product. Gleason-type assumptions are hypotheses on measures over projectors. Preparation calibration is additional input that ties a pure state to certainty on its own ray. Equivalence of projected

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realizations is a comparison of complete packages of sector, metric, observables, and dynamics, and it can fail even when each package is separately admissible [7]. A probability measure over projection labels is not constructed.

4.3. What Is Not Implied

Projection idempotency alone does not imply a unique physical metric, and a compatible metric does not imply Born's rule without the measure and calibration hypotheses just listed. Norm conservation of a G-unitary evolution is a global quadratic statement. It becomes a local continuity equation only after a local expression for the density and current has been derived from the dynamical operator, as in the Schrödinger illustration of Section 5.6. The companion paper records the same separation for its own hypotheses and does not claim an assumption-free derivation of Born's rule [7]. This manuscript uses that separation and does not add a derivation.

4.4. Role of the Present Section

The section is sufficient for the foundation paper if it keeps the dependency in the right order and refuses the shorter arrows. It is not a substitute for the companion theorems on operational equivalence, spectral obstructions, or the finite-dimensional criteria under a full matrix observable algebra. Those remain results of [7], valid for the realizations supplied there, and they are not premises from which a geometric PTQ model is constructed in this paper.

5. Projected Geometric Flow and Observable Dynamics

Dynamics are admitted, in this framework, only as evolution of a sector that has already been declared observable. The question of what evolves physically is not answered by unrestricted kinematical motion. No universal field equation is fixed here. What can be fixed is the class of constraints that a candidate evolution must satisfy before this paper will call it a physical law of PTQ.

5.1. Sector Preservation

A physical evolution law must act on, or induce a closed action on, the admissible projected sector. A larger underlying motion may be studied kinematically. It acquires the interpretation of this framework only through its induced action on states that remain in the sector. If a configuration is physical because it lies in the image of the projection, time development that leaves the image is formal kinematics, not a physical evolution law in the sense of OBP. This is a requirement imposed on candidate laws. It is not a theorem that every microscopically defined evolution preserves a given image.

5.2. Metric Compatibility

The physical inner product is part of the declared data of the sector. A candidate law is compatible with that choice only if it preserves the corresponding norm on the domain where the generator is self-adjoint with respect to the chosen product, or if any failure of preservation is stated as a breakdown of that interpretation. A law that moves the observables out of the declared geometry is not using the same notion of physicality that made the observables meaningful. Compatibility is an extra condition on the pair consisting of the metric and the generator. It is not implied by the existence of a projector.

5.3. Residual Non-Observable Structure

Projected dynamics do not make the full kinematical arena dispensable. The larger arena is the source of the observable sector and may contain couplings that influence the projected evolution. Those residual components are not reportable observables merely because they occur in the formulation. They can affect the admissible projection, modify effective geometric data on the sector, or constrain which projected flows stay inside the sector. Their mediating role is not a license to count every background degree of freedom as an additional observable mode.

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A published scalar-operator example makes a related, and sharper, distinction. Strict projection discards the complementary block, whereas exact Schur reduction lets that block contribute before it is eliminated, and the retained operators need not agree [8]. Residual structure can therefore change the retained dynamics even when probabilities are assigned only in the retained sector. That statement is native to the scalar Green-function setting of the cited paper. It is not a universal constitutive-closure theorem, and it is not a quaternionic spacetime theorem.

5.4. Admissible Form of Geometric Evolution

An evolution law is admissible in the present sense only if it meets four conditions. It is defined on, or induces a closed action on, the projected sector. It preserves the declared observability conditions along the motion. It is compatible with the chosen positive metric. It respects the continuity statement that has actually been derived for that law, rather than a continuity statement inferred from normalization alone.

These conditions exclude arbitrary model building. They are weaker than a regime-specific closure. They do not decide whether the effective evolution is a transport equation, a projected connection law, or a constrained flow on a bundle. Different regimes may realize the conditions differently. Agreement on the four conditions is not agreement on one equation, and it is not evidence that one residual geometry already fits several observational channels.

5.5. Interpretation

The order used by OBP is the reverse of attaching observability after the fact to unrestricted motion. The sector is declared; a complex structure and a positive metric are chosen compatibly; probability hypotheses, if used, are stated; and only then is a dynamical law admitted as physically meaningful in this framework. Projected geometric flow is that order, not a claim that the technical details of a gravitational field equation are already fixed.

The present section does not re-prove tensorial identities, does not derive a general transport system, and does not claim an effective equation valid in all regimes. Its role is to define what "dynamics" means after OBP has fixed the observable sector and after probability has been restricted to the conditional status of Section 4.

5.6. A Minimal Toy Model

The preceding sections can be illustrated by a minimal toy realization. The example is not a universal realization of PTQ, not a quaternionic field theory, and not a quantum-gravity construction. The space $\mathcal{K} \cong \mathbb{C}^2$ is used only as a four-real-dimensional representation in which an internal PT-type involution and its even image can be written explicitly. The involution acts on the internal coordinates (u, v). It is not shown to be a physical spacetime transformation of (x, t).

Consider

$$\mathcal{K} \cong \mathbb{C}^2, \qquad \Psi = \begin{pmatrix} u \\ v \end{pmatrix}. \tag{1}$$

Let parity exchange the components and let time reversal act by the ambient complex conjugation,

$$P\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v \\ u \end{pmatrix}, \qquad T\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u^* \\ v^* \end{pmatrix}, \tag{2}$$

so that

$$R\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v^* \\ u^* \end{pmatrix}. \tag{3}$$

The map R is an internal real-linear involution: $R^2 = \mathrm{Id}$, and R does not act on a spacetime point (x, t). In this subsection the symbol PT means this same internal map. It is an internal PT-type involution, not a demonstrated spacetime parity–time transformation.

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The ambient complex structure and the complex structure used for the physical sector are different real-linear maps:

$$I_0\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}iu\\iv\end{pmatrix},\qquad J\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}iu\\-iv\end{pmatrix}. \tag{4}$$

Direct evaluation gives $J^2 = I_0^2 = -\mathrm{Id}$. They are not scalar multiples of one another, since $J(0,1) = (0,-i)$ and $I_0(0,1) = (0,i)$. A real-linear combination $a\,\mathrm{Id} + bJ$, with $a, b \in \mathbb{R}$, acts by

$$(a+ib)\cdot_J\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}(a+ib)u\\(a-ib)v\end{pmatrix}. \tag{5}$$

This is ordinary complex scaling on the first component and the complex-conjugate scaling on the second. It is not the ambient multiplication I_0 unless $b = 0$ or v scales in the same way as u.

The PT eigenspaces are found from

$$\phi=\frac{u+v^*}{2},\qquad \chi=\frac{u-v^*}{2}, \tag{6}$$

which invert to $u = \phi + \chi$ and $v = \phi^* - \chi^*$. Thus

$$\Psi=\Psi_{\mathrm{phys}}+\Psi_{\mathrm{res}}, \tag{7}$$

with

$$\Psi_{\mathrm{phys}}=\begin{pmatrix}\phi\\\phi^*\end{pmatrix},\qquad \Psi_{\mathrm{res}}=\begin{pmatrix}\chi\\-\chi^*\end{pmatrix}, \tag{8}$$

and

$$PT\,\Psi_{\mathrm{phys}}=\Psi_{\mathrm{phys}},\qquad PT\,\Psi_{\mathrm{res}}=-\Psi_{\mathrm{res}}. \tag{9}$$

The physical projection is the real-linear map

$$\Pi_{\mathrm{phys}}\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}(u+v^*)/2\\(u^*+v)/2\end{pmatrix}. \tag{10}$$

It is idempotent, and its image is

$$\mathcal{H}_{\mathrm{obs}}=\left\{\begin{pmatrix}\phi\\\phi^*\end{pmatrix}:\phi\in\mathbb{C}\right\}. \tag{11}$$

The complementary image is the PT-odd set of vectors Ψ_{res}. Neither image is a complex subspace for I_0. If $\Psi = (\phi, \phi^*)$ lies in $\mathcal{H}_{\mathrm{obs}}$ and $\phi \neq 0$, then

$$I_0\Psi=\begin{pmatrix}i\phi\\i\phi^*\end{pmatrix},\qquad (i\phi)^*=-i\phi^*\neq i\phi^*, \tag{12}$$

so $I_0\Psi \notin \mathcal{H}_{\mathrm{obs}}$. The same vector is J-invariant in the image:

$$J\begin{pmatrix}\phi\\\phi^*\end{pmatrix}=\begin{pmatrix}i\phi\\-i\phi^*\end{pmatrix}=\begin{pmatrix}i\phi\\(i\phi)^*\end{pmatrix}. \tag{13}$$

The identification $\iota(\phi) = (\phi, \phi^*)$ therefore intertwines ordinary multiplication by i on the amplitude with J on $\mathcal{H}_{\mathrm{obs}}$. Relative to J, the observable sector is a complex line. Relative to I_0, it is only a real two-plane.

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Proposition 1 (Toy-model complex structure). *Let $\mathcal{K}$, I_0, J, PT, and Π_{phys} be as above, and define*

$$\langle \Psi_1, \Psi_2 \rangle_J = \frac{1}{2}(u_1^* u_2 + v_1 v_2^*). \tag{14}$$

Then:

1. $J^2 = -\mathrm{Id}$ *and* $I_0^2 = -\mathrm{Id}$, *while* $I_0 \neq \pm J$.
2. Π_{phys} *is real-linear and idempotent, commutes with* J, *and does not commute with* I_0.
3. $\mathcal{H}_{\text{obs}}$ *is invariant under* J *and under real scalars, and is not invariant under* I_0.
4. *The form* (14) *is positive definite and Hermitian for* J. *It is not sesquilinear for* I_0.
5. *If* $\iota(\phi) = (\phi, \phi^*)$, *then* $\langle \iota(\phi_1), \iota(\phi_2) \rangle_J = \phi_1^* \phi_2$.
6. PT *commutes with* J, *preserves* (14), *and is anti-linear with respect to* I_0.

Proof. Item (i) is the evaluation recorded after (4). Real-linearity of Π_{phys} holds because ambient conjugation is real-linear, so (10) preserves real linear combinations. If $\phi = (u + v^*)/2$, a second application replaces (u, v) by (ϕ, ϕ^*) and returns the same ϕ, hence $\Pi_{\text{phys}}^2 = \Pi_{\text{phys}}$. For commutation with J, the ϕ-coordinate of $J\Psi = (iu, -iv)$ is

$$\frac{iu + (-iv)^*}{2} = \frac{iu + iv^*}{2} = i\phi, \tag{15}$$

hence $\Pi_{\text{phys}} J\Psi = (i\phi, -i\phi^*) = J\Pi_{\text{phys}}\Psi$. The maps do not commute with I_0: $\Pi_{\text{phys}}(1,0) = (1/2, 1/2)$, so

$$\Pi_{\text{phys}} I_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} i/2 \\ -i/2 \end{pmatrix}, \qquad I_0 \Pi_{\text{phys}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} i/2 \\ i/2 \end{pmatrix}. \tag{16}$$

For item (iii), $J(\phi, \phi^*) = (i\phi, (i\phi)^*)$ lies in $\mathcal{H}_{\text{obs}}$, while $I_0(\phi, \phi^*) = (i\phi, i\phi^*)$ does not whenever $\phi \neq 0$, because $(i\phi)^* = -i\phi^*$.

Hermitian symmetry is the conjugation

$$\overline{\langle \Psi_2, \Psi_1 \rangle_J} = \overline{\frac{1}{2}\left(u_2^* u_1 + v_2 v_1^*\right)} = \frac{1}{2}(u_1^* u_2 + v_1 v_2^*) = \langle \Psi_1, \Psi_2 \rangle_J. \tag{17}$$

Positive definiteness is $\langle \Psi, \Psi \rangle_J = \frac{1}{2}(|u|^2 + |v|^2)$. For J-linearity in the second slot, $J\Psi_2 = (iu_2, -iv_2)$ and $(-iv_2)^* = iv_2^*$, so

$$\langle \Psi_1, J\Psi_2 \rangle_J = \frac{1}{2}(u_1^*(iu_2) + v_1(iv_2^*)) = i\langle \Psi_1, \Psi_2 \rangle_J. \tag{18}$$

The first slot is conjugate-linear:

$$\langle J\Psi_1, \Psi_2 \rangle_J = \frac{1}{2}((iu_1)^* u_2 + (-iv_1)v_2^*) = \frac{1}{2}(-iu_1^* u_2 - iv_1 v_2^*) = -i\langle \Psi_1, \Psi_2 \rangle_J. \tag{19}$$

The general real combination (5) therefore acts by the scalar $a + ib$ in the second slot. The form is not I_0-sesquilinear. For $\Psi_1 = \Psi_2 = (1, 1)$,

$$\langle \Psi_1, I_0 \Psi_2 \rangle_J = 0, \qquad i\langle \Psi_1, \Psi_2 \rangle_J = i. \tag{20}$$

Equivalently, the I_0-sesquilinear polarization of the same quadratic norm would replace $v_1 v_2^*$ by $v_1^* v_2$. On $\mathcal{H}_{\text{obs}}$ that alternative restricts to $\mathrm{Re}(\phi_1^* \phi_2)$, not to $\phi_1^* \phi_2$.

On $\mathcal{H}_{\text{obs}}$, $v_a = \phi_a^*$, so each term in (14) equals $\frac{1}{2}\phi_1^* \phi_2$ and the sum equals $\phi_1^* \phi_2$. Thus $\iota(i\phi) = J\iota(\phi)$ and the restricted product is the standard Hermitian product of the amplitude.

Finally, $PT(iu, -iv) = (iv^*, -iu^*) = J(v^*, u^*)$, so PT commutes with J. The same formula with I_0 gives $PTI_0 = -I_0 PT$. Substituting $PT\Psi = (v^*, u^*)$ into (14) returns the original value, and $PT^2 = \mathrm{Id}$. □

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Proposition 1 does not assert that (14) is the only positive product compatible with Π_{phys}. Idempotency was used to identify the image, not to derive the product. Define the internal fiber to be the image of the projection,

$$H_{\text{int}} := \mathcal{H}_{\text{obs}} = \Pi_{\text{phys}}(\mathcal{K}). \tag{21}$$

Relative to J, H_{int} is the complex line already identified by ι, so

$$\dim_{\mathbb{C}} H_{\text{int}} = 1. \tag{22}$$

The proposition concerns only this fiber. On that one-dimensional line, every positive J-Hermitian product is a positive multiple of $\phi_1^*\phi_2$. The multiple used here is the convention $\langle \iota(\phi), \iota(\phi)\rangle_J = |\phi|^2$. The fiber norm of $\iota(\phi)$ is the number $|\phi|^2$. There is no spatial integral in that norm. The scale is a convention on this toy line, not a uniqueness theorem on a larger observable algebra, and the companion probability analysis treats metric nonuniqueness and projection nonuniqueness as different freedoms [7]. The same formula (14) is one positive J-Hermitian product on the ambient four-real-dimensional space; other products exist there. No step identifies this finite-dimensional linear algebra with a quaternionic quantum field theory.

The Schrödinger illustration below is not a statement about a vector in H_{int}. It treats ϕ as a spatial wavefunction and takes the state space to be

$$H_{\text{wave}} := L^2(\mathbb{R}, H_{\text{int}}) \cong L^2(\mathbb{R}), \qquad \dim_{\mathbb{C}} H_{\text{wave}} = \infty. \tag{23}$$

A minimal dynamics on that amplitude is the additional illustrative law

$$i\partial_t \phi = \left(-\frac{1}{2m}\partial_x^2 + V(x)\right)\phi, \tag{24}$$

with $m > 0$. Two readings of this illustration are kept separate, and neither is a new dynamical law. The formal local continuity identity below assumes that V is real-valued and that $\phi(\cdot, t)$ is a C^2 solution for which the indicated derivatives exist pointwise. The unitary-evolution reading on H_{wave} needs one further restriction, which reality of V alone does not give: if V is real and essentially bounded, then $-\frac{1}{2m}\partial_x^2 + V$ is self-adjoint on the Sobolev domain $H^2(\mathbb{R}) \subset H_{\text{wave}}$, as a bounded symmetric perturbation of the free Laplacian, and the unitary group it generates conserves the L^2 norm. Less regular potentials are not treated. The factor i in (24) is the complex structure transported to the fiber amplitude by ι, hence the structure J on H_{int}, not a claim that H_{int} is invariant under I_0. If $\phi(\cdot, t)$ is a C^2 solution and V is real, reality of $-\partial_x^2$ gives $\partial_t \phi^* = (\partial_t \phi)^*$. For each fixed x, the value $\iota(\phi(x, t))$ therefore remains in H_{int}. The state itself, when it lies in H_{wave}, is the map $x \mapsto \iota(\phi(x, t))$, not a single vector of H_{int}. The internal involution R is not applied to (x, t) in this equation.

Define $\rho = |\phi|^2$ and

$$j = \frac{1}{m}\operatorname{Im}(\phi^*\partial_x\phi). \tag{25}$$

These are tied to (24), not to $\Pi_{\text{phys}}^2 = \Pi_{\text{phys}}$. Indeed

$$\partial_t \phi = \frac{i}{2m}\partial_x^2\phi - iV\phi, \tag{26}$$

and reality of V cancels the potential terms in $\partial_t \rho = (\partial_t \phi)^*\phi + \phi^*\partial_t\phi$, leaving

$$\partial_t \rho = \frac{i}{2m}\left(\phi^*\partial_x^2\phi - (\partial_x^2\phi)^*\phi\right) = \frac{i}{2m}\partial_x(\phi^*\partial_x\phi - (\partial_x\phi)^*\phi). \tag{27}$$

The parenthesis equals $2i\operatorname{Im}(\phi^*\partial_x\phi)$, so

$$\partial_t \rho + \partial_x j = 0. \tag{28}$$

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The second-derivative term is essential to the rearrangement. Equation (28) is that formal local identity. It is not the statement that every L^2 orbit satisfies a pointwise continuity equation. The spatial norm on H_{wave} is

$$\int_{\mathbb{R}} |\phi(x)|^2 \, \mathrm{d}x. \tag{29}$$

Integrating (28) conserves (29) only after the endpoint contribution of j is shown to vanish, as it does for Schwartz-class data. Under the bounded-potential hypothesis above, conservation of (29) also follows from unitarity on L^2 and does not require that pointwise rearrangement. The integral is not the fiber norm. The fiber norm of $\iota(\phi)$ in H_{int} is the number $|\phi|^2$, with no spatial variable. A J-unitary group on H_{int} conserves the fiber norm and defines no spatial current. Unitarity on H_{wave} conserves (29) and still does not, by itself, provide the local current j.

The density $\rho = |\phi|^2$ is the quadratic density of the fiber product, evaluated on the wavefunction. Reading it as a probability density would require measure hypotheses that are not established in this manuscript. Gleason's complex-dimension hypothesis, as used by the companion transport, fails for H_{int} because $\dim_{\mathbb{C}} H_{\text{int}} = 1$. The same dimension hypothesis is satisfied for H_{wave} because that space is infinite-dimensional [7]. Satisfaction of the dimension count is not a derivation. The measure, noncontextuality, and preparation hypotheses required by that transport are not established here for either space. No Born-rule derivation is claimed for the fiber or for the wavefunction illustration, and no probability distribution over a family of projections is constructed.

The proposition shows only that an internal checklist is non-empty on H_{int}: an internal PT-type involution, a complex structure that leaves the fiber invariant, and one compatible positive product whose fiber norm is $|\phi|^2$. The Schrödinger illustration is a separate statement on H_{wave}. Its local current is derived from the Schrödinger operator, not from idempotency of Π_{phys} and not from a spacetime action of R. Geometric residual content is not calculated here.

6. Observable Consequences and Falsifiability

A framework that restricts dynamics by sector preservation, metric compatibility, and a derived continuity statement can still fail. The failure conditions below are requirements and exclusion tests. They are not reports of a completed empirical identification. Effective-field-theory and modified-gravity languages are used only as the comparison vocabulary already standard in the cited reviews [5,6]. This section does not fit data and does not claim empirical success.

If a deviation is to be attributed to PTQ, OBP asks that it be interpretable as an effect of the residual structure left by a declared projection, inside the admissibility conditions of Sections 2–5. That is an observational filter on attributions. It is not a proof that a unique constitutive model exists, and it is not a proof that every observed deviation has this origin. A phenomenological template, even one motivated by the same spurion language, is not thereby a uniquely predicted observable.

6.1. Channels in Which a Test Could Be Posed

Geometry, transport, and consistency conditions are tested through dynamical behavior. Cosmological evolution, structure growth, galactic dynamics, gravitational-wave propagation, and quantum-sector norm and continuity are channels in which a declared residual structure might be confronted. Listing a channel does not assert that a distinguishable PTQ signal has been derived for it. No measurement survey is given here.

6.2. Framework-Level Expectations, Not Unique Predictions

At the level of principle, OBP does not instruct one to insert an unrelated substance in each regime. The proposal is that a deviation attributed to the program should be expressible from the declared projected geometry, should remain compatible with the metric and continuity statements actually proved for that realization, and should not require mutually contradictory residual structures in sectors that are asserted to be the same realization.

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These are expectations imposed on future constructions. They do not by themselves predict the sign or the scale of a deviation. They do not select a unique operator, a unique numerical template, or a universal stress-energy correction. Model-dependent formulas in companion preprints remain model-dependent until an independent measurement interface identifies them.

6.3. Failure Conditions

Failure is indexed to a specified realization and to the assumptions that realization declares. Three cases stay separate.

Realization admissibility fails when that realization cannot exhibit, together, a projected sector, a complex structure on that sector, a positive compatible metric, an observable representation, and the dynamics it intends to use. The failure is mathematical inadmissibility of the declared package.

Empirical failure of a particular model is a separate case. It occurs when a prediction that model states, under a measurement map it states, is contradicted, or when the stated comparison does not yield an observable distinguishable from the benchmark in that comparison. The public record used here has not established a PTQ-specific discriminating prediction. That absence limits the empirical support for PTQ as a distinct theory. It does not empty a single-channel admissible realization merely because that realization states no cross-channel relation.

Cross-regime closure fails when one specified realization claims shared physical content across regimes and then misses a cross-regime relation it has declared. The failure constrains that claim under the hypothesis combination that was tested. A model-level failure of this kind is not a framework-wide no-go, and it does not transfer to a companion theorem that did not enter the tested combination. A realization that makes no cross-channel claim is not emptied by the lack of such a claim.

Deviations obtained only by leaving the declared sector, by dropping a continuity statement the dynamics was required to satisfy, or by adding compensating components unrelated to the declared projection, count against that realization. None of these conditions is an observational exclusion of every possible realization.

6.4. A Model-Dependent Cross-Scale Stress Test

A cross-regime identity is required only where one specified realization explicitly claims applicability in more than one regime. It is not a defining test of every PTQ branch.

The public SPARC diagnostic is a stress test of one such claim [9]. It uses the weak-field matching $a_0 = \varepsilon\, c\, H_0$ as a selected relation, not as a unique PTQ prediction, and it uses the cosmological map $\Omega_\Lambda(\varepsilon) = \varepsilon^2/(1+\varepsilon^2)$ as an adopted diagnostic, not as a derived cosmological law. Under that conjunction, strict cross-scale equality fails: $\varepsilon_{\mathrm{RC}} \approx 0.155$ and $\varepsilon_{\cos} \approx 1.492$. No independent dimensionless geometric bridge is established. The kinematic diagnostic in that preprint is not identified with a dimensionless geometric efficiency, and no such efficiency is introduced here to close the gap. On the common mapped domain, the matched PTQ-ν and MOND simple-ν likelihood families are observationally identical, so the comparison does not identify a PTQ-specific radial law. The SPARC sample is not a validation of complete quaternionic gravity. The negative result stands for the hypothesis combination that was tested. It is not a universal no-go for PTQ, and it is not assigned to the tensor, scalar, probability, or trace-residue results cited elsewhere in this paper.

A separate preprint organizes cosmological, galactic, and collider profiles as templates that depend on scale, in a minimal-alignment posture, and states that those profiles are not one universal equation [10]. Templates of that kind are not unique predictions of the principles in Section 2.

A success confined to one sector would not validate a realization that claims several regimes. Failure of this selected equality removes the claim that the tested realization has already achieved cross-scale closure. It leaves room for a future realization stated under different assumptions, and it does not supply the missing empirical discrimination.

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6.5. Interpretation

Observable deviations are not introduced here as new autonomous substances. If they are attributed to PTQ, they are to be read as questions about a declared projected structure. The section does not say that open problems in cosmology, galactic dynamics, or quantum consistency have been solved. The attribution rule can fail in the three ways just separated. The cited diagnostic has not closed the cross-regime claim it tests, and the program has not established an independent empirical discrimination.

7. Relation to Public PTQ Results

This section locates the manuscript among public results. It does not rank them, and it does not assert that they are consequences of one theorem proved here.

7.1. Role of the Present Manuscript

This paper states OBP, the four organizing principles, the conditional status of probability and current, the constraint form of observable dynamics, and one finite-dimensional illustration of the complex structure on a PT-even image. It is not a specialized theorem paper, not a regime-specific model, and not an empirical fit. Later conditional constructions can be compared with OBP. They do not become theorems of this paper by being cited.

7.2. Conditional Results Used as Scope Markers

The public record relevant to that comparison is the following. Each item is limited to the statement actually made in the cited version.

1. The Symmetry paper proves three tensor-sector statements under explicit assumptions (A1)–(A6): algebraic reduction of torsion to a pure-trace form aligned with a compensator gradient in that paper's convention, a bulk equivalence among three routes modulo improvements, and a coefficient lock $K = G$ for tensor modes on the admitted domains, hence luminal tensor speed at that quadratic order, with a stated next-to-leading correction [11]. Those assumptions include a two-derivative truncation and a spurion limit in which the compensator is not varied. The result is not a derivation of the principles of the present paper, and it is not a completed gravitational theory.
2. The Particles paper constructs the Green function of a real scalar on a fixed real Lorentzian background. Quaternionic data enter only through a fixed auxiliary two-component representation of a principal-symbol deformation, which is not a spacetime metric. Strict projection retains the diagonal block, while exact Schur reduction retains an additional complementary contribution, and the two operators need not agree. No quaternionic spacetime geometry, no full interacting quaternionic quantum field theory, and no observational result is derived [8].
3. The probability preprint studies when different supplied admissible projected realizations represent the same physical probability theory, and which invariants obstruct that equivalence. It assumes a complex sector, a represented observable algebra, a retained generator, and a positive compatible metric. It does not classify geometric projections, does not prove metric uniqueness, and does not put a probability measure on projection labels [7].
4. The projective Palatini preprint is a local conditional classification in a compensated trace-residue class. Under its last-index projective law and its compensator convention, strict removal of the retained residue holds on the stated local branch if and only if f_2 vanishes identically and f_3' vanishes identically. The compensator is a nondynamical spurion and is not varied. The result is not an unrestricted metric-affine theorem and not a prohibition on every projective formulation [12]. The compensator convention of [11] is a different variational definition and is not identified with this one.
5. Further public branches remain conditional and separate. A UV/D-brane preprint gives one branch-level route from selected D3-brane data to a projected observable sector; it is not a

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universal string derivation of PTQ [13]. The SPARC preprint is a weak-field diagnostic with the negative cross-scale statement already cited [9]. The dark-sector preprint is a template-level phenomenology, not a universal field equation [10]. A further preprint constructs a conditional scalar-gradient trace-torsion branch in a PT-even Einstein–Cartan setting and does not claim a galactic fit, a dark-sector replacement, or a cosmological completion [14]. These papers are not assembled here into a single effective theory.

7.3. Distinct Reduction Operations

The word "projection" covers several operations that this program must not identify. Table 1 records the distinction at the level needed for the present scope. The table is a glossary, not a new theorem.

Table 1. Reduction operations that remain distinct. Each row is limited to the public source named in Section 7 or to Proposition 1.

Operation	Mathematical character	Not the same as
Anti-linear microscopic PT	Anti-linear relative to the complex structure for which conjugation is defined. In the toy model this anti-linearity is with respect to I_0, while the same map is J-linear.	The internal grading Θ, and not automatically a state projector.
Real-linear PT-even projection	The toy map Π_{phys}: real-linear, idempotent, J-linear, and not I_0-linear. Image $\mathcal{H}_{\text{obs}}$.	A complex-linear projection for the ambient structure I_0.
Complex-linear grading Θ	In the scalar-operator paper, a linear grading that separates a retained component from a complementary component [8].	Microscopic anti-linear PT. The two are distinguished in that paper.
Strict sector restriction	Discard the complementary block and retain the diagonal operator.	Exact elimination. The retained operators differ when the complementary coupling is kept [8].
Schur/Feshbach reduction	Eliminate the complementary block after its coupling has contributed. Standard as a method; the cited result is the sign and the retained scalar operator in that deformation.	Strict restriction, and not a universal Hamiltonian theorem.
Projective gauge quotient	In the Palatini trace-sector preprint, a quotient of an extended trace-plus-compensator space under a stated projective law. The compensator is not a varied field [12].	Π_{phys}, strict scalar projection, and the compensator convention of the Symmetry paper.

7.4. Division of Labor with the Probability Paper

Reference [7] is independent of the toy-model proposition. It takes admissible realizations as supplied and compares their probability theories. The present paper does not use those equivalence theorems as a derivation of OBP, and it does not enlarge them into a geometric existence theorem. Conversely, nothing proved here about J and Π_{phys} is a substitute for the companion's obstruction criteria.

8. Discussion

8.1. What Is Standard, and What OBP Adds

Several ingredients used above are standard and are not results of this paper. Pseudo-Hermitian and PT-symmetric quantum mechanics already separate an auxiliary inner product from a positive physical metric and already know that the metric need not be fixed by the generator alone [1–4]. Projection-operator and Schur–Feshbach reductions are standard elimination methods; the scalar paper's contribution is their controlled comparison for one auxiliary deformation, not the invention of block elimination [8]. Projective invariance of the trace sector in Palatini and metric-affine gravity is

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standard; the cited classification is local and conditional [12]. Constrained quantization, superselection, and gauge quotient constructions are likewise standard ways of refusing to treat every kinematical variable as an observable. This paper does not rederive them.

What the paper adds is the decision to organize PTQ by OBP, together with the negative statements required to keep that decision honest. Observable status is attached to a declared sector before a probability rule is discussed. The physical complex structure is distinguished from the ambient one when they differ, as they do in the toy model. Metric choice, measure hypotheses, and local transport are listed as further inputs. Companion constructions are accepted only inside their stated hypotheses and are not promoted to a single PTQ theorem. The toy-model proposition is the one statement proved in full in these pages, and its scope is the finite-dimensional representation in which it is proved.

What remains a choice of formalism is the adoption of OBP itself. Nothing proved here shows that a generalized quantum theory is inconsistent unless it selects observables by a PT-even projection of quaternionic kinematics. A theory may use constraint quantization, a superselection rule, or an operator-level C operator and never mention OBP. PTQ, as written here, chooses projection-defined physicality as its organizing principle. That choice is not a law deduced from the companions.

8.2. Scope

The paper therefore establishes a framework of interpretation and one illustrative proposition. It does not provide a cosmological model, a complete account of structure formation, or a finished weak-field realization. It does not replace quantum theory. It does not derive a universal dynamical system, and it does not settle classification, uniqueness, or effective closure.

8.3. Limitations and Open Problems

The classification and possible non-uniqueness of admissible projections remain open. The four requirements in Section 2 are not a construction procedure. Proposition 1 shows that at least one finite-dimensional PT-even image can be equipped with a complex structure that leaves it invariant. It is not a classification theorem, and it says nothing about observer-dependence.

The positive metric remains an input. Even the restricted uniqueness-up-to-scale statement in the toy model uses the fact that the J-complex dimension is one. A larger represented observable algebra can change the freedom, and this paper does not prove a general fixing theorem.

No closed dynamical system is given for all regimes. Section 5 states constraints on candidate laws. Existence, uniqueness, and comparative adequacy of concrete realizations are open. In particular, local elimination of an auxiliary field in one model is not a universal constitutive closure.

Cross-regime predictive closure is not established for realizations that claim it. Section 6 records one model-level failure of a declared cross-scale equality; the comparison is not repeated here. That limitation binds models that assert a shared cross-regime structure. It is not a necessary condition on every independent mathematical result in the program, and it does not refute a companion theorem that does not assert the tested map. Independent empirical discrimination of a PTQ residual, as distinct from compatibility of a template with a data set, remains an open limit on support for PTQ as a distinct theory.

Any ultraviolet completion is unresolved. The D-brane preprint is one conditional branch construction and is not that completion [13].

These limitations are the main fronts on which the program can mature or fail: projection classification, metric uniqueness, dynamical completion, and independent empirical identification.

8.4. Relation to the Other Public Branches

The probability paper develops the equivalence theory that Section 4 only delimits [7]. The weak-field preprint tests dictionaries and reports a cross-scale mismatch under its own map [9]. The template preprint and the conditional geometric preprints provide separate arenas [10,12,14]. They are accountable to the principles only in the weak sense that a proposed observable reading should say

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which sector, which metric, and which reduction it is using. They are not evidence that the principles have already survived those tests.

8.5. Outlook

PTQ, as presented here, is a framework with an explicit organizing principle and an explicit toy-model distinction between two complex structures. It is not a finished theory. Further work can classify projections, constrain metrics by stated observable algebras, and construct dynamics whose continuity laws are derived rather than inferred. It can also fail the tests in Section 6. Enlarging the interpretive claims of this paper is not a substitute for those tasks.

9. Conclusions

OBP organizes PTQ by declaring observable physicality only on an admissible projected sector. Once that declaration is made, a complex structure, a compatible positive metric, probability-measure hypotheses, and compatible dynamics are further inputs. They are not implied by idempotency of a projector. Under those inputs the framework can carry a conditional probability interpretation and a continuity statement whose local form has actually been derived. The toy-model proposition realizes the fiber checklist on the one-dimensional internal space H_{int}, with complex structure J rather than the ambient structure I_0. The illustrative Schrödinger current is a statement on H_{wave} and is derived from the Schrödinger operator, not from a spacetime action of the internal involution.

The distinction from neighboring subjects should stay narrow. Ordinary quantum theory may take a probability interpretation as primitive. PT-symmetric and pseudo-Hermitian quantum mechanics may organize admissibility through operators and a metric. Modified-gravity model building may carry a deviation by an extra effective component. OBP proposes a different order for PTQ. The proposal does not refute those subjects, and it does not show that they are special cases of one projection theorem.

Three questions remain open, and they are not one question. The first is realization admissibility and construction: whether a declared projection, complex structure, positive metric, observable representation, and dynamics can be exhibited together. The second is independent empirical discrimination: whether a specified realization predicts an observable distinguishable from existing benchmarks. No such PTQ-specific discrimination is established in the public record used here. The third is model-dependent cross-regime closure: whether a realization that claims shared physical content across regimes satisfies the cross-regime relations it declares. Cross-scale equality belongs to that third question. It is not a defining test of every PTQ branch. The weak-field diagnostic fails the selected equality it states, and the template preprint does not supply a universal equation [9,10].

The manuscript does not reproduce the tensor-sector proofs, the scalar Schur analysis, the probability-equivalence theorems, or the local trace-residue classification [7,8,11,12]. It does not convert conditional ultraviolet, weak-field, or geometric branches into a single theory [9,10,13,14]. Its role is to state the organizing principle under which those constructions can be interpreted and compared, and under which a realization can be set aside when a failure condition applies to that realization.

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